Class_17_Hw

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```r
my_data <- read.table("asthma_results.txt")

library(ggplot2)

summary(my_data)
```

```
    sample              geno                exp        
 Length:462         Length:462         Min.   : 6.675  
 Class :character   Class :character   1st Qu.:20.004  
 Mode  :character   Mode  :character   Median :25.116  
                                       Mean   :25.640  
                                       3rd Qu.:30.779  
                                       Max.   :51.518  
```

```r
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

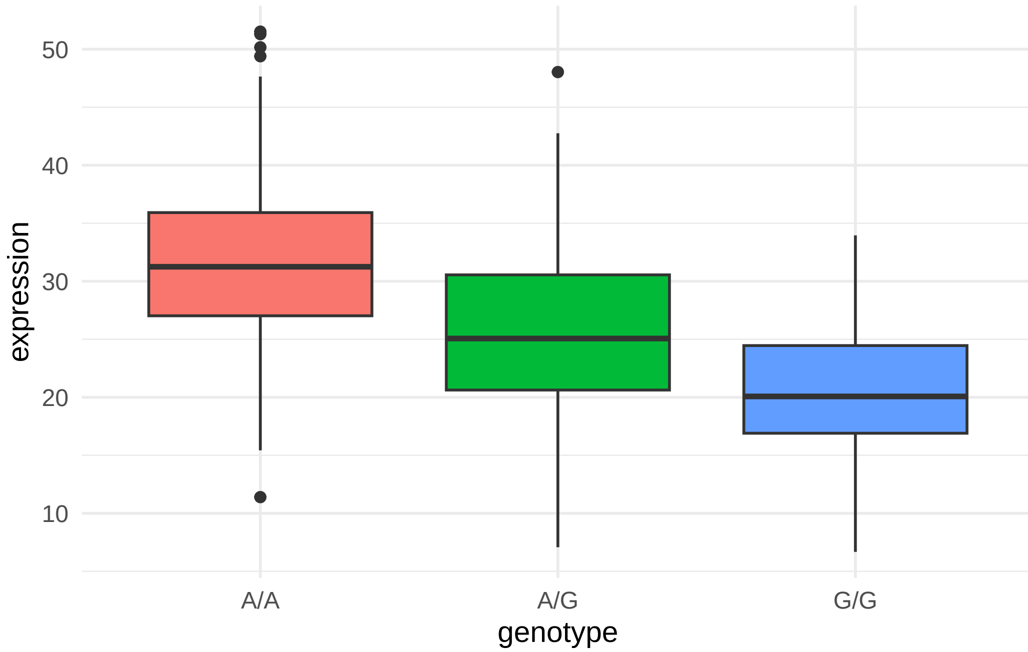
```
AA <- my_data %>% filter(geno == "A/A")

AG <- my_data %>% filter(geno == "A/G")

GG <- my_data %>% filter(geno == "G/G")
```

> Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

The A/A genotype has 108 samples with a median expression of 31.248475. The A/G genotype has 233 samples with a median expression of 25.06486.The G/G genotype has 121 samples with a median expression of 20.07363.

> Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
my_plot <- ggplot(my_data) +
  aes(x = geno, y = exp,
      fill = geno) +
  geom_boxplot() +
  labs(x = "genotype", y = "expression") +
  theme_minimal() +
  theme(legend.position = "FALSE")
my_plot
```



The SNP does effect expression levels with A/A showing the highest expression. Between AA and GG there is about a 12 point decrease in expression. These results indicate that the SNP allele of ORMDL3 can cause a change in expression levels of the protein product wich can lead to and increased risk of childhood asthma.